

Statistics & Probability



Mean, median, and mode

- 1 Find the mean of 4, 8, 6, 10, 7.
 - 2 Find the median of 3, 9, 5, 1, 7.
 - 3 Find the mode of 2, 4, 4, 6, 8, 4, 2.
 - 4 A data set has values 8, 12, 10, 15, 10, 14, 11. Find the mean, median, and mode.
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Range and quartiles

- 5 Find the range of 5, 12, 8, 20, 3.
 - 6 For the data set 2, 4, 5, 7, 8, 9, 12, find Q_1 and Q_3 .
 - 7 Find the interquartile range (IQR) for the same data set.
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Variance and standard deviation

- 8 Find the variance of 2, 4, 6, 8, 10, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.
 - 9 Find the standard deviation of the same data set, rounded to two decimal places.
 - 10 A data set has variance 25. State its standard deviation.
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Frequency tables

- 11 A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.
 - 12 Using the same frequency table, find the mode.
 - 13 Using the same frequency table, find the median.
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Basic probability

- 14 A fair six-sided die is rolled. Find $P(\text{number} > 4)$.
- 15 A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.
- 16 A card is drawn from a standard 52-card deck. Find $P(\text{king})$.
- 17 Two fair coins are flipped. Find $P(\text{both heads})$.
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Complementary and mutually exclusive events

- 18 If $P(A) = 0.35$, find $P(\text{not } A)$.
- 19 A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?
- 20 If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.
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Independent events

- 21 Two independent events have $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
- 22 A fair coin is flipped twice. Find $P(\text{first heads and second tails})$.
- 23 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are aces})$.
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Conditional probability

- 24 In a class, $P(\text{math}) = 0.7$ and $P(\text{physics and math}) = 0.4$. Find $P(\text{physics} \mid \text{math})$.
- 25 A box has 10 balls: 6 red and 4 blue. Two balls are drawn without replacement. Find $P(\text{2nd is blue} \mid \text{1st is blue})$.
- 26 Given $P(A) = 0.5$, $P(B) = 0.3$, $P(A \text{ and } B) = 0.15$, find $P(A \mid B)$.
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Counting: permutations and combinations

- 27 In how many ways can 4 books be arranged on a shelf?

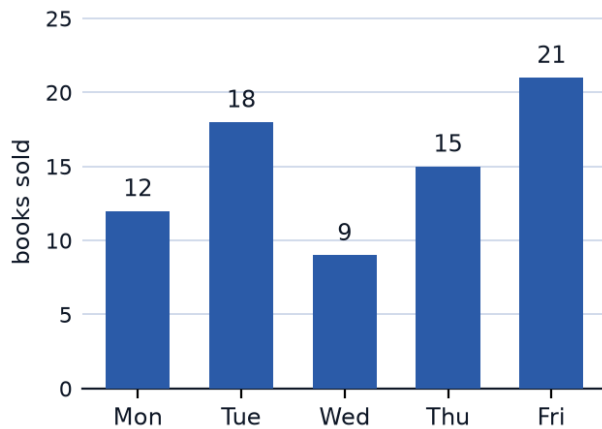
- 28 How many ways can a committee of 3 be chosen from 8 people (order doesn't matter)?
- 29 A code uses 3 distinct digits from 0–9, where order matters. How many possible codes are there?
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The five-number summary

- 30 For the data set 6, 10, 14, 18, 22, 26, 30, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).
- 31 Find the IQR for the same data set.
- 32 Outlier boundaries are $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$. Find both boundaries for the same data set.
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Reading data from a bar chart

- 33 The bar chart shows the number of books sold each day at a bookstore over five days.



- a Find the total number of books sold over the five days.
- b Find the mean number of books sold per day.
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Word problems: statistics and probability in context

- 34 This question has two parts. A survey of 200 students found that 120 like math and 90 like science, while 50 like both subjects.
- a Find the probability that a randomly selected student likes math or science, using $P(M \cup S) = P(M) + P(S) - P(M \cap S)$.
- b Find the probability that a student likes neither subject.

- 35** This question has two parts. A quality-control inspector finds that 2% of items from a production line are defective. A random sample of 3 items is tested, and defects occur independently.
- a** Find the probability that all three items are defective.
 - b** Find the probability that none of the three items are defective.